

SHUDI WENG

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EDUCATION

KTH Royal Institute of Technology, Stockholm, Sweden

2023 – Exp. 2027

PH.D. IN ELECTRICAL ENGINEERING.

Information Science and Engineering (ISE), Electrical Engineering Department

Advisors: Prof. Mikael Skoglund and Prof. Ming Xiao

Technical University Berlin, Berlin, Germany

Oct 2025 – Jan 2026

RESEARCH VISIT

The Theoretical Foundations of Communications Technology (CommIT) Group

Advisors: Prof. Giuseppe Caire

Chalmers University of Technology, Gothenburg, Sweden

2020 – 2022

M.SC. IN INFORMATION AND COMMUNICATION TECHNOLOGY.

Communication System Group, Electrical Engineering Department

Advisor: Prof. Henk Wymeersch

Beihang University, Beijing, China

2016 – 2020

B.SC. IN ELECTRICAL ENGINEERING.

Electrical Engineering Department

Minor: Control Technology, Optics

Peter the Great St.Petersburg Polytechnic University, St.Petersburg, Russia

2018

EXCHANGE STUDENT.

Courses: Optoelectronics, Russian

RESEARCH INTERESTS

Federated Learning, Privacy, Secure Aggregation, Information Theory and Coding.

PUBLICATIONS

JOURNAL

- [1] **Shudi Weng**, Xiang Zhang, Yizhou Zhao, Giuseppe Caire, Ming Xiao, Mikael Skoglund. “Dropout-Resilient and Communication-Efficient Secure Aggregation in Semi-decentralized Federated Learning”. In: *In Preparation for International Symposium on Information Theory (ISIT)*, 2025.
- [2] **Shudi Weng**, Chao Ren, Yizhou Zhao, Ming Xiao, Mikael Skoglund. “Secure Cooperative Gradient Coding”. In: *Submitted to IEEE Journal on Selected Areas in Communications (JSAC)*, 2025.
- [3] **Shudi Weng**, Chao Ren, Ming Xiao, Mikael Skoglund. “Heterogeneity-Aware Client Sampling: A Unified Solution for Consistent Federated Learning”. In: *Submitted to IEEE Transactions on Signal Processing (TSP)*, 2025.
- [4] **Shudi Weng**, Chao Ren, Ming Xiao, Mikael Skoglund. “Cooperative Gradient Coding”. In: *Accepted by IEEE Transactions on Communications (TCOM)*, 2025.
- [5] **Shudi Weng**, Ming Xiao, Chao Ren, Mikael Skoglund. “Coded Cooperative Networks for Semi-Decentralized Federated Learning”. In: *IEEE Wireless Communications Letters (WCL)*, 2024.

- [6] **Shudi Weng**, Fan Jiang, Henk Wymeersch. “Wideband mmWave Massive MIMO Channel Estimation and Localization”. In: *IEEE Wireless Communications Letters (WCL)*, 2023.

CONFERENCE

- [1] **Shudi Weng**, Chengxi Li, Ming Xiao, Mikael Skoglund. “Cooperative Gradient Coding for Semi-Decentralized Federated Learning”. In: *IEEE Global Communications Conference (Globecom)*, 2024.

EXPERIENCE

RadChat, Gothenburg, Sweden

Sep 2022 – Dec 2022

CORE SYSTEM DEVELOPER.

Prototyped a RIS-aided joint radar and communication system, including system design, software verification, and hardware implementation

Top 5 to join CampX accelerator by Volvo Group

Swedish Mining Innovation Idea Competition Winners 2022.

China Airborne Missile Academy, Henan, China

Summer 2019

INTERN.

Participated and learned the missile manufacturing process

SERVICE

Reviewer, Journal & Conference

2023-Present .

IEEE Journal on Selected Areas in Communications (JSAC)

Transactions on Communications (TCOM)

Conference on Neural Information Processing Systems (NeurIPS)

IEEE Global Communications Conference 2026 (GLOBECOM)

IEEE International Conference on Communications (ICC)

Information Theory Workshop (ITW)

Teaching, KTH

2023 – 2026

FEO3250/EQ2845 INFORMATION THEORY AND SOURCE CODING.

Specialization course for M.Sc. and Ph.D students in electrical engineering and computer science

Led 12 tutorial lectures and assessed assignment and exam grading

Organizing Committee, Conference & Workshops

2026 .

Sustainable Cyber-Physical Software-Defined System Slicing (SUCCESS) Workshop

Supervision,

2026 .

Josef Afreim, Master's Thesis, Weather Radar Methods for Portable Military Radar System (with Saab)

Co-Organizer, KTH

March 2023 – Sep 2025

DIVISION OF INFORMATION SCIENCE AND ENGINEERING (ISE).

Weekly Research Seminars in Communication Theory and Information Theory for Approx. 40-70 ppl.

TALKS

Invited Talk at Decentralized Learning Workshop, Linköping University, Sweden

Jun 2026

Titled: *Fundamental Limits of Resilient and Efficient Hierarchical Secure Aggregation.*

Invited Talk at SUCCESS workshop, KTH, Sweden

Mar 2026

Titled: *Fundamental Limits of Resilient and Efficient Hierarchical Secure Aggregation.*

Invited Talk at TU Berlin on Decentralized AI, Germany

Oct 2025

Titled: *Coding for Federated Learning.*

Invited Talk at LiU-KTH-Ericsson Decentralized AI Inspiration Workshop, Sweden

Sep 2025

Titled: *Secure and Resilient Coding for Federated Learning.*

AWARDS & GRANTS

Travel Grant from General Travel Foundations , KTH, Sweden	<i>2026</i>
Travel Grant from EECS school's foundations , KTH, Sweden	<i>2025</i>
Travel Grant from Karl Engvers Foundation , KTH, Sweden	<i>2024</i>
International Scholarships , Chalmers, Sweden	<i>2021-2022</i>
Best Bachelor Thesis , Beihang University, China	<i>2020</i>
First-Class Scholarship , Beihang University, China	<i>2019</i>
China Scholarship Council (CSC) Scholarship , Ministry of Education of China	<i>2018</i>
Special-Class Scholarship , Beihang University, China	<i>2018</i>

SKILLS

Python, Pytorch, Tensor Flow, Matlab, C, HTML, Ubuntu, L^AT_EX

LANGUAGES

English (Professional), Chinese (Native Speaker), Swedish (Elementary), Russian (A2 in 2019)

OTHER INTERESTS

Photography, Colored Pencil Drawing